Data Science

Lecture 07: Data Exploration and Preprocessing

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Where We Left Off

Three lectures fitted models to tables nobody questioned. Today steps back to KDD steps 1–2 from Lecture 01: what the columns are, whether they can be trusted, and what a plot shows that a summary hides.

Lecture 01
data is never raw

→

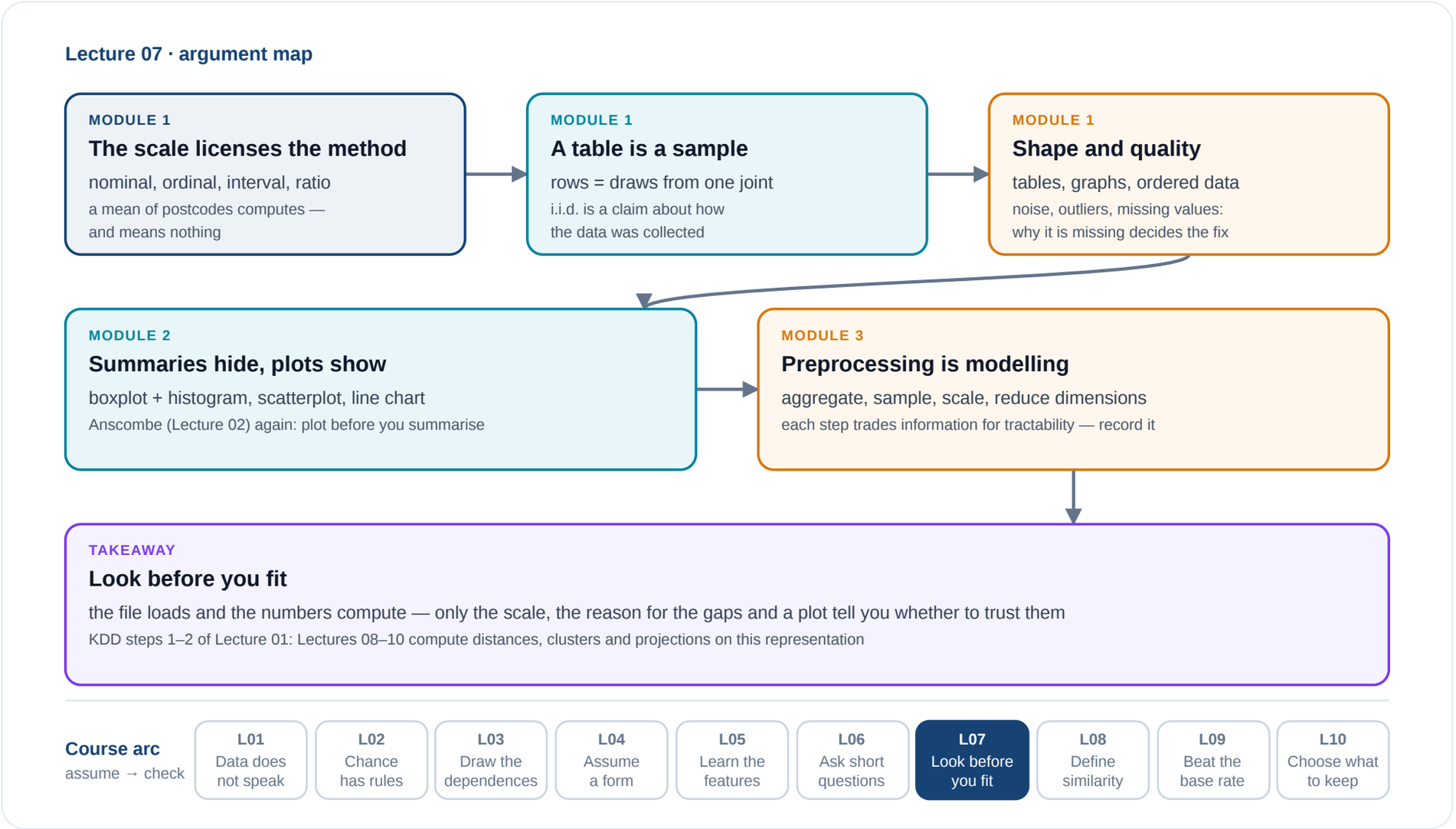
Lecture 02
summaries can agree while the data differs (Anscombe)

→

Today
scale, quality, plots, preprocessing

> From Lecture 08 on there is no label to check a model against — looking first stops being good practice and becomes the main check we have.

Argument Map



Module 1

What Kind of Data Is This?

> ***Guiding Question:*** *The file loads and the numbers compute. Why is that not yet a reason to trust the average?*

Four Attribute Types — What the Values Allow

	type	the values…	meaningful	statistics	examples
categorical	**nominal**	are names that only distinguish objects	$=, \neq$	mode, entropy, contingency table, χ^2 test	ZIP code, employee ID, eye colour
	ordinal	carry an order	$<, >$	median, percentiles, rank correlation, sign test	grades, rankings, {good, better, best}
numeric	**interval**	have meaningful differences; the zero is arbitrary	$+, -$	mean, standard deviation, Pearson correlation, t-test	temperature in °C or °F, calendar dates
	ratio	have a true zero, so ratios are meaningful	$\times, \div$	geometric mean, harmonic mean, coefficient of variation	length, mass, counts, age, temperature in K

Each type permits everything the one above it does, plus more.

Four Attribute Types — What You May Transform

	type	permissible transformation	example
categorical	**nominal**	any one-to-one relabelling	reassigning employee IDs changes nothing
	ordinal	any strictly increasing f: $x \mapsto f(x)$	{good, better, best} is as well coded by {1, 2, 3} as by {1, 5, 9}
numeric	**interval**	$x \mapsto ax + b$, $a > 0$	°C and °F differ in zero point and unit
	ratio	$x \mapsto ax$, $a > 0$	length in metres or in feet

INTUITION

A statement is **meaningful** on a scale if no permissible transformation can change its truth. "Twice as long" survives metres → feet; "twice as warm" does not survive °C → °F.

Why the Type Decides the Method

- ✗ **The classic error:** averaging a nominal attribute. The mean of ZIP codes 10115 and 81667 is a number, and it is meaningless.
- ✗ **The subtle one:** 20 °C is not "twice as warm" as 10 °C — interval scales have no true zero, so ratios do not transfer.

> The attribute type determines which **statistics**, which **distance measures** and which **plots** are permissible. Every method in this course depends on it.

A Table Is a Sample From a Joint Distribution

	payment	satisfaction	born	income (k€)	returned?
column	X_1 · nominal	X_2 · ordinal	X_3 · interval	X_4 · ratio	X_5 · nominal
draw $\mathbf{x}_1$	card	4	1987	52	yes
draw $\mathbf{x}_2$	invoice	2	1995	31	no

Each **column** is a random variable **with a measurement scale**; each **row** is one draw $\mathbf{x}_i$ from **one joint distribution** $p(x_1, \ldots, x_5)$ that nobody has written down.

A is 180 cm long, B 90 cm. Is A "twice as long"?

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length recorded as	A	B	what the values support
cm — **ratio**	180	90	"twice as long" — in inches 70.9 vs 35.4, still ×2
cm above 100 cm — **interval**	80	−10	"90 cm longer"; the ratio −8 means nothing
size class 1–3 — **ordinal**	3	1	"longer"; "three times" means nothing
size label — **nominal**	L	S	"different" — nothing more

The Scale Suggests the Model

Which distribution would you write down for each column — and how many numbers does it need?

> The scale limits the model as it limits the statistics: a normal distribution for ZIP codes is as meaningless as their mean.

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attribute	values	scale	a first model	numbers
returned?	yes / no	nominal	**Bernoulli**(p)	1
payment	card, invoice, … (k)	nominal	**categorical**$(p_1, \ldots, p_k)$	$k - 1$
satisfaction	1 – 5 stars	ordinal	categorical (order unused)	4
orders last month	0, 1, 2, …	ratio, count	**Poisson**(λ)	1
height	around 170 cm	ratio	**normal**(μ, σ^2)	2
income	> 0, skewed	ratio	**log-normal**: normal on $\log$ income	2
share of orders returned	$[0, 1]$	ratio	**Beta**(α, β)	2

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What "i.i.d." Claims

The rows $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are **independent and identically distributed** (i.i.d.) if

independent	$p(\mathbf{x}_1, \ldots, \mathbf{x}_n) = \prod_i p_i(\mathbf{x}_i)$	*one customer's row tells nothing about another's*
identically distributed	$p_i = p$ for every row i	*every row comes from the **same** joint distribution*

Which of these datasets break it: daily sales of one shop · two visits by the same customer · pixels of one image · a survey sent to a random sample?

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Only the random survey is plausibly i.i.d. Time series, repeated measurements and spatial data all violate it — the arrangement *is* information.

Four Shapes of Data

Record data

Objects × attributes — the familiar table. Rows are *treated as* independent.

Graph data

Objects carry **links** to other objects: web pages, social networks, citations.

Fields / grids

Objects sit in a **spatial** arrangement: images, sensor lattices, simulations.

Spatio-temporal

Position **and** time: monthly land temperatures, traffic movement, trajectories.

> The shape decides which methods even apply. A table whose rows are treated as independent can be shuffled without loss; a grid or a time series cannot — the arrangement *is* information. Independence of rows is an assumption, not a property of the file.

Where Does a Recorded Value Come From?

A sensor records 22.4 °C. Besides the temperature, what went into that number?

$$x = s + b + \varepsilon$$

s: the true value

b: artefact — deterministic, the same every time

ε: noise — random, mean 0

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> A dataset records x, never s. Every data problem is a claim about the terms that were added on the way.

Three Problems on One Axis



Which of the three problems does collecting more data fix?

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Only **noise**. An artefact shifts every new reading too; an outlier comes back in the same proportion.

Each Problem Is a Model Statement

problem	in the model	more data	what helps
noise	$\varepsilon \sim p(\varepsilon)$, $E[\varepsilon] = 0$	averages it out	model it; average
artefact	$x = g(s)$, deterministic	does **not** help	calibration, a second instrument, domain knowledge
outlier	a draw from another q	a share π remains	robust statistics, or model q: $(1-\pi)\,p + \pi\,q$
missing	M = missing, $P(M \mid x)$	depends on why	ask for the mechanism first

None of these can be read off the data alone. Calling a value noise, an artefact or an outlier is an assumption about how it was generated.

Mini-Exercise: Stars and Birth Years

A customer table records satisfaction (one to five stars) and year of birth. Which scale is each — and whose **mean** would you not report?

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Satisfaction is **ordinal**: its mean assumes equal gaps between the stars — report the median. Year of birth is **interval**: the mean year is fine, but ratios are not ("born twice as late" means nothing); differences such as age are fine.

Mini-Exercise: Three Scales

Patients are weighed on three scales, a third on each. Two record $X = S + \varepsilon$; the third is miscalibrated and adds exactly 3 kg every time: $X = S + 3 + \varepsilon$. Does weighing ten times as many patients bring the mean of X closer to $E[S]$?

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> No. $E[X] = E[S] + \frac{1}{3} \cdot 3 = E[S] + 1$ kg: the mean converges, but to a value 1 kg too high — an **artefact**. More data shrinks only the noise ε.

Mini-Exercise: A Sensor That Fails in the Cold

Temperature T, missingness indicator M (1 = reading missing). The sensor fails mostly in the cold: $P(M = 1 \mid T)$ is large for low T. After deleting the incomplete rows, does the mean estimate $E[T]$ — or what?

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> It estimates $E[T \mid M = 0]$, which is **larger** than $E[T]$: the coldest days are gone. Only if M is **independent** of T is $E[T \mid M = 0] = E[T]$ and deletion harmless.

Takeaway

> A dataset is n **draws from an unknown joint distribution** — and every way of reading it is an assumption about how it was generated.

1. **Shape:** only record data can plausibly be read as independent rows.
2. **Table:** columns are random variables, rows are draws from $p(x_1, \ldots, x_d)$.
3. **Scale:** licenses a column's statistics and suggests its model.
4. **i.i.d.:** lets the rows factorise — to be checked, not presumed.
5. **Recorded value:** truth + artefact + noise. More data removes **noise only**; artefacts, outliers and missing values need a model of how they arose.

The Churn Shop's Customer Table

customer	contract	satisfaction	tenure (months)	postcode	churned?
1	monthly	4	3	94032	yes
2	yearly	—	27	80331	yes
3	yearly	5	41	94032	no
…	…	…	…	…	…

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SCALES

Contract, postcode, churned: nominal. Satisfaction: ordinal. Tenure: ratio. The mean postcode computes — and means nothing.

WHY IS SATISFACTION MISSING?

Customers who left stop answering: 40 of 100 churners gave no rating, 45 of 900 stayers. Drop those rows, and churn falls from **10%** to **6.6%**.

Module 2

Seeing the Data

> ***Guiding Question:*** *Summary statistics answer the questions you thought to ask. What shows you the ones you did not?*

Visualization Is Everywhere

Newspaper tables, subway maps, weather charts, stock graphs, a 3-D reconstruction of an injured knee, demographic bar charts, social-network layouts.

What do these have in common? All of them present data in a form the eye processes faster than the equivalent numbers.



Why It Works

Bandwidth

Vision is the human sense with the highest data rate — pattern detection runs in parallel and pre-attentively.

Interaction

A picture is a natural surface for exploring: zoom, filter, select, compare.

Discovery

Visualization surfaces the **unexpected** — the structure nobody thought to compute a statistic for.

Summary statistics answer questions you already asked. A plot lets the data ask you one — which is exactly what Anscombe's quartet demonstrated.

Visualization is not decoration for results; it is an **instrument of analysis**, and the only one that can show you what you did not think to measure.

The Boxplot

A **visual summary of a distribution built from quartiles**:

The box

Spans Q_1 to Q_3 — the interquartile range, the middle half of the data.

The line

The **median** inside the box. Its position shows skew.

The whiskers

(Tukey) Reach to the most extreme data points inside $[Q_1 - 1.5 \cdot \mathrm{IQR}, Q_3 + 1.5 \cdot \mathrm{IQR}]$. Points outside are drawn singly as **outliers**.

The 1.5 × IQR rule is a **convention**, not a test. It flags candidates for inspection, not errors for deletion.

The Histogram

What it shows

The **shape** of a distribution: modality, skew, gaps and clusters that a boxplot hides entirely.

What it hides

Everything depends on the **bin width**. Too wide and structure vanishes; too narrow and noise looks like structure.

A bimodal distribution and a unimodal one can produce the same boxplot. Always look at one histogram before trusting a five-number summary.

Mini-Exercise

Two datasets have the same minimum, Q_1, median, Q_3 and maximum — so the same boxplot. Must their histograms look alike?

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> No. The five numbers fix five points; the mass between them can be arranged freely — spread evenly inside the box, or piled into two clusters at its edges. Same quartiles, opposite stories: Anscombe's quartet again.

Mini-Exercise

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No. The five numbers fix five points; the mass between them can be arranged freely — spread evenly inside the box, or piled into two clusters at its edges. Same quartiles, opposite stories: Anscombe's quartet again.

A boxplot compares distributions compactly; a histogram shows their shape. **Neither alone is sufficient**, and the boxplot's outlier rule is a convention you may overrule.

The Scatterplot

An X–Y plot of two dimensions against each other — the standard instrument for inspecting a **relationship** rather than a distribution.

With more than two dimensions there are four options:

Subsetting

Let the user choose which dimensions to display.

Reduction

Project to fewer dimensions (see preprocessing, and projections later).

Embedding

Map extra dimensions to other channels: colour, size, shape.

Multiple displays

Superimpose, or juxtapose in a **scatterplot matrix**.

What a Scatterplot Reveals

Linear association, non-linear structure, clusters, outliers and heteroscedasticity — most of which no single correlation coefficient reports.

Pearson correlation $r = \dfrac{\operatorname{cov}(X,Y)}{s_X s_Y} \in [-1,1]$: the covariance measured in units of the two standard deviations. It measures **linear** association only.

A correlation of 0 is consistent with a perfect parabola. The scatterplot shows it; the coefficient does not.

Tufte's Principle

Maximise the data-ink ratio: every drop of ink on the page should carry information. Delete anything that does not — heavy gridlines, boxes, redundant tick marks, decorative fills. **Range frames** apply the principle to the axes: draw each axis only over the interval the data actually spans, so the frame itself reports the minimum and maximum.

> The exercise: take a finished chart and ask of each element, *what would I lose by deleting this?*

A scatterplot shows the structure a coefficient summarises away. Beyond two dimensions you must **subset, reduce, embed or juxtapose** — and every mark you draw should earn its ink.

When a Line Is Justified

Connecting points with a line asserts an **order** between them — and often an interpolation.

> **Data requirement:** one or more quantitative value attributes on a comparable scale, plus one **ordered** key attribute (usually time).

> Connecting nominal categories with a line claims an order that does not exist. The line is a statement, not a decoration.

Aspect Ratio Is Not Cosmetic

Readers compare trends by the **orientation** of the line segments, so the shape of the plotting area changes what they conclude. The same series looks flat in a wide frame and dramatic in a tall one.

Banking to 45° chooses the aspect ratio so that the segments' average absolute orientation is 45°: differences in orientation are judged best near ±45° (Cleveland et al. 1988).

Additional channels

Colour and line type distinguish series — but only for a handful of them.

Multiple Series

Overlaid

Best for comparing a few series directly against each other.

Stacked

Each series is drawn on top of the previous one. Shows the total well — and makes individual series **hard to read**, since each sits on a moving baseline.

Sorted / small multiples

One panel per series, ordered by some property. Scales to many series where overlaying fails.

A line asserts order and interpolation; the **aspect ratio** decides what the reader sees and stacking trades the readability of the parts for the readability of the whole.

Module 3

Preprocessing

> ***Guiding Question:*** *Lecture 01 claimed that preprocessing is modelling. Which choices made here change what a model can learn?*

Aggregation

Combine several objects or attributes into one — customer purchases summarised by store, by month, by price band.

Gains

Fewer objects to process; **less variation**, hence more stable estimates; the natural operation of a data warehouse.

Costs

Detail is gone, irreversibly. Effects that exist only within a group become invisible.

Sampling

Work with a subset of the objects: random sampling with or without replacement, or **stratified** sampling that preserves group proportions.
Sampling always loses information. The question is whether it loses anything that matters — a sufficiently large sample retains the overall structure.

> Sample a sine wave once per period and every sample has the same value: the reconstruction is a flat line — not merely rougher, but **wrong**.

Scaling and Encoding

Scaling numeric attributes

Min–max: $x' = \dfrac{x - \min}{\max - \min} \in [0, 1]$

Standardisation: $z = \dfrac{x - \bar{x}}{s}$ — mean 0, standard deviation 1

Encoding categorical attributes

Nominal → one-hot: one 0/1 column per value, so no false order.

Ordinal → ranks $1, 2, 3, \ldots$, which keep the order.

Two customers differ by 1,000 € in income and 30 years in age. Unscaled, their distance is $\approx 1{,}000$: age is invisible. Standardised ($s = 10{,}000$ € and 15 years) the differences are 0.1 and 2 — now age dominates.

Dimensionality Reduction

The curse of dimensionality. As the number of attributes grows, the data becomes sparse: distances between points concentrate, neighbourhoods lose meaning, and the volume to be covered grows exponentially.

The problem

Many features are irrelevant, noisy or correlated.

The cost

More computation, more parameters, more variance — and weaker distance-based methods.

The response

Feature selection keeps a subset; **feature extraction** builds new combined dimensions.

Every preprocessing step **trades information for tractability**. The trade is usually worth it — but it is a modelling decision, and it should be recorded as one.

Image Credits & References

Cleveland, William S. and McGill, Marylyn E. and McGill, Robert (1988). The Shape Parameter of a Two-Variable Graph. *Journal of the American Statistical Association*.